

Career Platform

Step-by-step deployment guide for beginners

Frontend (Next.js) and backend (FastAPI) on one AWS EC2 server, with PostgreSQL, and public access over HTTPS (Caddy with a Let's Encrypt certificate).

Domain	career.qualcsi.com
Repository	github.com/qulacsi-Tech/career-exploration-lead
Server	AWS EC2, Ubuntu 24.04 LTS, t3.medium, region ap-south-1 (Mumbai)
Purpose	Testing and demonstration. Read section 2 before inviting anyone.

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1. How the system fits together

There are four running parts on one server. You only need to know what each one does.

Part	What it is	Who reaches it
Internet (ports 80 and 443)	Visitors reach the server directly. Ports 80 and 443 are open; everything else is closed.	Anyone on the internet
Caddy	A small web router. It sends <code>/api/...</code> requests to the backend and everything else to the frontend.	Visitors, on ports 80 and 443
Frontend (Next.js)	The public website and the admin panel.	Only Caddy, inside the server
Backend (FastAPI) and database (PostgreSQL)	The API, and the data store.	Only Caddy and the backend, inside the server. The database is never exposed.

Visitors type `https://career.qualcsi.com`. The request reaches the server on port 443. Caddy checks the address and sends it to the frontend or to the backend. Caddy also gets the HTTPS certificate from Let's Encrypt, and renews it automatically.

Why this layout? Only SSH (for you), HTTP and HTTPS are open. The database and backend are never reachable directly from the internet. Caddy handles the certificate, so you do not manage it by hand.

2. Before you start

2.1 What you need

- An AWS account with billing enabled. You need permission to create EC2 instances and Elastic IPs.
- Control of the DNS for **qualcsi.com**, so you can add one record (Part B).
- Read access to the GitHub repository `qualacsi-Tech/career-exploration-lead`.
- A computer with Windows, macOS or Linux, and an SSH tool. Windows 10 and 11 include `ssh` in PowerShell.
- A password manager, for the secrets you will create in Part F.

2.2 Cost

Item	Approximate monthly cost (USD)
EC2 t3.medium, running all the time	about 30
Storage, 30 GB	about 3
Elastic IP (while attached to a running instance it is free; it is billed when the instance is stopped)	about 0 to 4
DNS and HTTPS certificate (Let's Encrypt)	0

Stop the instance when nobody is testing, to save money. Section M explains how.

2.3 Decisions already made

Decision	Value used in this guide
Region	ap-south-1 (Mumbai)
Instance	t3.medium, Ubuntu 24.04 LTS
Database	PostgreSQL 16, inside the server, in a container
Public access	Caddy with HTTPS (Let's Encrypt); ports 80 and 443 open

Important: this is a test server. The sample content is not real data. Do not send real student information through it. Read section O before sharing the address with anyone.

Part A. Create the AWS items

Use the AWS website: **console.aws.amazon.com**. Sign in, then follow each step in order.

A1. Choose the region

Step 1.

In the top-right corner of the AWS console, click the region name. Choose **Asia Pacific (Mumbai) ap-south-1**.

Every item in this part must be created in the same region. If you change region later, the items will not be visible.

A2. Create a key pair (your login key for the server)

Step 2.

Search for **EC2** at the top, and open it. In the left menu, click **Key pairs**, then **Create key pair**.

- Name: `career-demo`
- Key pair type: **ED25519**
- Private key file format: **.pem**

Click **Create key pair**. Your browser downloads `career-demo.pem`.

Keep this file private. Anyone with it can log in to your server. Do not email it or upload it anywhere. On Windows, move it to a folder such as `C:\Users\<you>\.ssh\`.

A3. Create a security group (the server's firewall rules)

Step 3.

In EC2, click **Security Groups** in the left menu, then **Create security group**.

- Security group name: `career-web`
- Description: `Demo server: SSH, HTTP, HTTPS`
- VPC: leave the default
- Inbound rules: click **Add rule**
 - Type: **SSH**, Port: 22, Source type: **My IP**
 - Type: **HTTP**, Port: 80, Source: **Anywhere** (0.0.0.0/0)
 - Type: **HTTPS**, Port: 443, Source: **Anywhere** (0.0.0.0/0)
- Outbound rules: leave the default (all traffic)

Click **Create security group**.

SSH is open only to your current address. HTTP and HTTPS are open to everyone, because visitors use them. If your internet address changes, edit the SSH rule (Security Groups → career-web → Inbound rules → Edit) and choose **My IP** again.

A4. Launch the server (EC2 instance)

Step 4.

In EC2, click **Instances**, then **Launch instances**.

Setting	What to choose
Name and tags: Name	career-demo
Application and OS Images (AMI)	Ubuntu , then Ubuntu Server 24.04 LTS (HVM), SSD Volume Type, 64-bit (x86)
Instance type	t3.medium (2 vCPU, 4 GiB memory)
Key pair (login)	career-demo
Network settings: Firewall (security groups)	Select existing security group , then career-web
Configure storage	30 GiB, gp3

Click **Launch instance**. Click **View all instances**. Wait until **Instance state** shows **Running** and **Status check** shows **2/2 checks passed** (about 2 minutes).

A5. Give the server a fixed address (Elastic IP)

Step 5.

Without this, the server's address changes every time it restarts.

1. In the left menu, click **Elastic IPs**, then **Allocate Elastic IP address**, then **Allocate**.
2. Select the new address in the list. Click **Actions**, then **Associate Elastic IP address**.
3. Resource type: **Instance**. Choose career-demo . Click **Associate**.
4. Write the address down. This guide calls it **EIP**. An example is 13.201.10.25 .

Check A: In **Instances**, career-demo shows **Running** and its **Public IPv4 address** equals your EIP.

Part B. Point the domain at the server

Create a DNS record that sends `career.qualcsi.com` to your server's Elastic IP. Caddy then gets the HTTPS certificate for it automatically.

B1. Create the DNS record

Step 6.

Sign in to the company that manages **qualcsi.com** (for example AWS Route 53, GoDaddy or Namecheap). Open its DNS settings and add this record:

Field	Value
Type	A
Name / Host	<code>career</code>
Value / Points to	your Elastic IP (EIP)
TTL	300 (or the default)

B2. Check the record

Step 7.

On your computer, open PowerShell and run:

```
nslookup career.qualcsi.com
```

Check B: The answer shows your Elastic IP. It can take 5 to 30 minutes after you save the record. Wait until it does before going on.

B3. Confirm the web ports

Step 8.

Ports 80 and 443 were opened in Part A3 (AWS) and are opened in Part D3 (server firewall). Confirm the AWS rules now: EC2 → Security Groups → `career-web` → Inbound rules shows SSH, HTTP 80 and HTTPS 443.

Part C. Connect to the server

Step 9.

On your computer, open **PowerShell** (Windows) or **Terminal** (macOS and Linux). Run the command below. Replace the two placeholders.

```
ssh -i C:\Users\<you>\.ssh\career-demo.pem ubuntu@EIP
```

The first time, you will see a question about the server's fingerprint. Type `yes` and press Enter.

When the prompt changes to `ubuntu@ip-...:~$`, you are on the server. Every command from here on runs on the server, unless the step says otherwise.

Check C: Type `whoami`. It prints `ubuntu`.

Mistakes to avoid. Use `Ctrl+Shift+V` or right-click to paste into the terminal. Copy each command exactly, including any capital letters. If a command shows a red error, read it before running the next one.

Part D. Prepare the server

D1. Update Ubuntu and install tools

Step 10.

```
sudo apt update && sudo apt -y full-upgrade
sudo apt -y install ca-certificates curl git ufw fail2ban unattended-upgrades
sudo dpkg-reconfigure -plow unattended-upgrades
```

In the last step, a blue screen asks about automatic updates. Choose **Yes**. Press Enter.

D2. Add swap memory

The website build needs more memory than the server has, so we add 4 GB of swap (disk used as extra memory).

Step 11.

```
sudo fallocate -l 4G /swapfile
sudo chmod 600 /swapfile
sudo mkswap /swapfile
sudo swapon /swapfile
echo '/swapfile none swap sw 0 0' | sudo tee -a /etc/fstab
free -h
```

Check D2: The line starting with `Swap:` shows `4.0Gi`.

D3. Firewall: allow SSH, HTTP and HTTPS

Step 12.

```
sudo ufw default deny incoming
sudo ufw default allow outgoing
sudo ufw allow OpenSSH
sudo ufw allow 80/tcp
sudo ufw allow 443/tcp
sudo ufw --force enable
sudo ufw status verbose
```

Check D3: The status lists **OpenSSH**, **80/tcp** and **443/tcp**, and nothing else allowed.

D4. Install Docker

Docker runs each part of the platform in its own sealed container, so setup is repeatable.

Step 13.

```
sudo install -m 0755 -d /etc/apt/keyrings
sudo curl -fsSL https://download.docker.com/linux/ubuntu/gpg -o /etc/apt/keyrings/docker.asc
sudo chmod a+r /etc/apt/keyrings/docker.asc
echo "deb [arch=$(dpkg --print-architecture) signed-by=/etc/apt/keyrings/docker.asc]
https://download.docker.com/linux/ubuntu $(. /etc/os-release && echo $VERSION_CODENAME) stable"
| sudo tee /etc/apt/sources.list.d/docker.list > /dev/null
sudo apt update
sudo apt -y install docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-
plugin
sudo usermod -aG docker ubuntu
```

Step 14.

Leave the server and connect again, so the new group membership applies:

```
exit
```

Reconnect with the same command as step 9. Then:

```
docker version
docker compose version
docker run --rm hello-world
```

Check D4: hello-world prints *Hello from Docker!*

Part E. Download the code

The repository is private. The server gets its own read-only key to download it.

Step 15. Create the server's key.

```
ssh-keygen -t ed25519 -C "career-demo-server" -f ~/.ssh/github_deploy -N ""  
cat ~/.ssh/github_deploy.pub
```

The last command prints one long line that starts with `ssh-ed25519`. Select it and copy it.

Step 16. Add the key to GitHub (on your computer, in a web browser).

1. Open `github.com/qulacsi-Tech/career-exploration-lead`.
2. Click **Settings** (top menu of the repository), then **Deploy keys** in the left menu.
3. Click **Add deploy key**. Title: `career-demo-server`. Paste the key. Leave **Allow write access** unticked.
4. Click **Add key**. You may be asked for your GitHub password.

Step 17. Clone the repository (back on the server).

```
cat >> ~/.ssh/config <<'EOF'  
Host github.com  
  IdentityFile ~/.ssh/github_deploy  
  IdentitiesOnly yes  
EOF  
chmod 600 ~/.ssh/config  
ssh-keyscan github.com >> ~/.ssh/known_hosts  
sudo mkdir -p /opt/career-platform  
sudo chown ubuntu:ubuntu /opt/career-platform  
git clone git@github.com:qulacsi-Tech/career-exploration-lead.git /opt/career-platform  
cd /opt/career-platform  
git log --oneline -1
```

Check E: The last line shows a commit (short code and a message). The code is on the server at `/opt/career-platform`.

Part F. Write the configuration and secrets

What is a secret? A password or key the system uses. Each one is random and long, and it is written only on the server. Never paste a secret into a chat, a document, a screenshot or a Git commit.

Step 18. Create three secret values.

Run the command three times. Each run prints a different random value. Copy each into your password manager, with a label.

```
openssl rand -hex 24      # label: POSTGRES_PASSWORD
openssl rand -hex 16      # label: ADMIN_PASSWORD (for the demo admin login)
openssl rand -hex 32      # label: SECRET_KEY
```

Step 19. Create the environment file.

```
cd /opt/career-platform/deploy
cp .env.example .env
nano .env
```

The `nano` editor opens. Use the arrow keys to move. Change these lines (keep the rest as it is):

```
DOMAIN=career.qualcsi.com
ACME_EMAIL=you@example.com
POSTGRES_PASSWORD=...first secret from step 18...
ADMIN_EMAIL=admin@career.qualcsi.com
ADMIN_PASSWORD=...second secret from step 18...
ADMIN_NAME="Demo Admin"
```

To save and close nano: press `Ctrl+O`, then `Enter`, then `Ctrl+X`.

Step 20. Create the backend file.

```
cp backend.env.example backend.env
nano backend.env
```

Change these two lines:

```
DATABASE_URL=postgresql+asyncpg://postgres:...first secret from step 18...@db:5432/leadgendb
SECRET_KEY=...third secret from step 18...
```

The password in `DATABASE_URL` must be exactly the same as `POSTGRES_PASSWORD`. If they differ, the backend cannot connect to the database.

Save and close nano, as before.

Step 21. Lock both files so only you can read them.

```
chmod 600 .env backend.env  
ls -l .env backend.env
```

Check F: Both lines start with `-rw-----`.

Part G. Start the database and the backend

Step 22. Start the first three containers.

```
cd /opt/career-platform/deploy
docker compose -f docker-compose.prod.yml --env-file .env up -d --build db backend
docker compose -f docker-compose.prod.yml ps
```

The first build takes 2 to 5 minutes. Wait until the list shows `db` with `(healthy)`. If it is still `starting`, wait 30 seconds and run the second command again.

Step 23. Check the backend answers.

```
docker compose -f docker-compose.prod.yml exec backend python -c "import
urllib.request;print(urllib.request.urlopen('http://localhost:8000/api/ping').read())"
```

Check G: The output is `b'{"status":"ok"}'`.

Part H. Create the tables and sample data

Run this part only once, on an empty database. Running the sample-data step twice creates duplicate colleges. If you need to start again, see section N.

Step 24. Create the database tables.

```
cd /opt/career-platform/deploy
set -a; . ./env; set +a
docker compose -f docker-compose.prod.yml --env-file .env run --rm backend alembic upgrade head
```

The output lists several lines ending in `Running upgrade ... -> 009`.

Step 25. Load the sample data and create the demo admin account.

```
docker compose -f docker-compose.prod.yml --env-file .env run --rm -e
ADMIN_EMAIL="$ADMIN_EMAIL" -e ADMIN_PASSWORD="$ADMIN_PASSWORD" -e ADMIN_NAME="$ADMIN_NAME"
backend python seed.py
```

Check H: The last line is `Seed complete`.

Part I. Start the proxy and the HTTPS certificate

Step 26. Start the router. On first start it requests the certificate from Let's Encrypt.

```
cd /opt/career-platform/deploy
docker compose -f docker-compose.prod.yml --env-file .env up -d caddy
docker compose -f docker-compose.prod.yml logs --tail=40 caddy
```

Look for a line containing `certificate obtained successfully`. It can take up to a minute. If you see challenge errors, check DNS (Part B), then ports 80 and 443 (Parts A3 and D3), then run `docker compose -f docker-compose.prod.yml restart caddy`.

Step 27. Check the address works from the server.

```
curl -s https://career.qualcsi.com/api/ping
```

Check I: The output is `{"status":"ok"}`. If it is not, see section N, "Certificate problems".

Part J. Build and start the website

The website is built with the public address, so the proxy and its certificate must be working first (Part I).

Step 28. Build the frontend. This takes 5 to 15 minutes. Do not close the window.

```
cd /opt/career-platform/deploy
docker compose -f docker-compose.prod.yml --env-file .env build frontend
```

Step 29. Start it and check every container.

```
docker compose -f docker-compose.prod.yml --env-file .env up -d frontend
docker compose -f docker-compose.prod.yml ps
```

Check J: Every row (backend , caddy , db , frontend) shows Up . The db row shows (healthy) .

Part K. Test the site

Do these checks from **your own computer**, not from the server. Use a private browser window for the admin checks.

#	Open this address	What you should see
1	<code>https://career.qualcsi.com</code>	The home page with the padlock.
2	<code>http://career.qualcsi.com</code>	It opens the HTTPS version (Caddy redirects).
3	<code>https://career.qualcsi.com/api/v1/home</code>	Text in curly brackets containing <code>"success": true</code> .
4	<code>https://career.qualcsi.com/courses</code>	A list of courses such as MBA and BBA.
5	<code>https://career.qualcsi.com/compare</code>	The comparison list.
6	<code>https://career.qualcsi.com/search?q=mba</code>	Matching colleges.
7	<code>https://career.qualcsi.com/sitemap.xml</code>	A page of XML addresses.
8	<code>https://career.qualcsi.com/exams/cat/practice</code>	Practice test cards. Open one, answer a question, press Submit, confirm: the result page appears.
9	<code>https://career.qualcsi.com/admin</code>	It opens the login page (this is correct).
10	Log in on <code>/login</code> with <code>admin@career.qualcsi.com</code> and the <code>ADMIN_PASSWORD</code> from step 18	The admin panel opens with its dashboard.
11	Admin → Content → Study Abroad. Change one word, press Save, open the public <code>/study-abroad</code> page, then change the word back.	The change shows on the public page. Your test confirms saving works.

Check K: All 11 checks pass. Then repeat checks 1, 4 and 8 on a phone using mobile data, which is a different network.

Part L. Close extra ports and set up backups

Only SSH, HTTP and HTTPS should be open. Nothing else.

Step 30. Confirm the security group rules.

In AWS: EC2 → Security Groups → `career-web` → Inbound rules. There must be three rules: SSH from **My IP**, HTTP 80 and HTTPS 443 from anywhere. Delete any other rule.

Step 31. Set up a daily backup at 02:30.

```
chmod +x /opt/career-platform/deploy/backup.sh
(crontab -l 2>/dev/null; echo '30 2 * * * /opt/career-platform/deploy/backup.sh >> /opt/career-
platform/deploy/backups/backup.log 2>&1') | crontab -
/opt/career-platform/deploy/backup.sh
ls -lh /opt/career-platform/deploy/backups/
```

Check L: The list shows a file named `leadgendb-....sql.gz`.

Copy backups off the server regularly. In the AWS console, the simplest way is to download files with an SFTP program such as WinSCP, which connects with the same key. A better option, for later, is to copy them to a private Amazon S3 bucket.

Part M. Everyday operations

M1. Look at the logs

```
cd /opt/career-platform/deploy
docker compose -f docker-compose.prod.yml logs --tail=100 backend
docker compose -f docker-compose.prod.yml logs --tail=100 frontend
docker compose -f docker-compose.prod.yml logs --tail=100 caddy
```

Add `-f` after `logs` to follow live. Press `Ctrl+C` to stop.

M2. Restart one part

```
docker compose -f docker-compose.prod.yml restart frontend
```

M3. Update to a new version

```
cd /opt/career-platform
git pull
cd deploy
docker compose -f docker-compose.prod.yml --env-file .env run --rm backend alembic upgrade head
docker compose -f docker-compose.prod.yml --env-file .env up -d --build backend frontend
```

The database changes are applied first, then the new code starts.

M4. Roll back to the previous version

```
cd /opt/career-platform
git log --oneline -5
git checkout <previous-commit-code>
cd deploy
docker compose -f docker-compose.prod.yml --env-file .env up -d --build backend frontend
```

Replace `<previous-commit-code>` with the code from the list (for example `3cb39d1`). Database changes are not undone automatically. If a change went wrong, restore the database from a backup (section N).

M5. Stop the demo to save money

Stop the server's containers first, then the instance:

```
cd /opt/career-platform/deploy
docker compose -f docker-compose.prod.yml --env-file .env stop
```

In the AWS console: EC2 → Instances → select `career-demo` → Instance state → **Stop instance**. The Elastic IP stays attached, so the address is the same when you start again.

To start again: EC2 → Instances → **Start instance**. Then connect with SSH and run:

```
cd /opt/career-platform/deploy
docker compose -f docker-compose.prod.yml --env-file .env up -d
```

M6. Check disk space

```
df -h /
docker system df
docker image prune -f
```

`docker image prune` removes old, unused images. It is safe.

Part N. When something goes wrong

Find your symptom in the first column. Try the fix, then check again. If it still fails, copy the log output (section M1) and ask for help with the step number.

What you see	Likely cause	Fix
SSH: <i>Permission denied (publickey)</i>	Wrong key file, or wrong user name	Use the <code>.pem</code> from step 2 and the user <code>ubuntu</code> . On Windows, the key file must not be shared with other users (right-click → Properties → Security).
SSH: <i>Connection timed out</i>	Your internet address changed	EC2 → Security Groups → career-web → edit the SSH rule → source My IP again.
Site shows 502 Bad Gateway	The frontend or backend container is down	<code>docker compose -f docker-compose.prod.yml ps</code> , then restart the one that is not Up.
<code>curl .../api/ping</code> does not work, but the containers are Up	DNS or the certificate is not ready	Check <code>nslookup career.qualcsi.com</code> , then the Caddy log (Part I).
Build of the frontend fails with <i>fetch failed</i>	Caddy was not ready with a certificate during the build	Complete Part I first, then run step 28 again.
<code>backend</code> logs show a database error	<code>POSTGRES_PASSWORD</code> and the password in <code>backend.env</code> do not match	Make them identical (step 20), then <code>up -d backend</code> .
Admin login says <i>Could not sign in</i>	Wrong password, or the sample-data step (26) did not run	Use the ADMIN_PASSWORD from step 18. Check the output of step 25.
Site shows a blank page or an error after a change	A stale build or a failed update	Run M3 again. Check <code>logs --tail=100 frontend</code> .
Build runs out of memory	Swap not active	<code>free -h</code> ; if Swap is 0, repeat step 11.
Disk full	Old images or backups	Section M6; delete old files in <code>deploy/backups/</code> after copying them.

N0. Certificate problems

Browser warnings about the certificate, or `challenge failed` in the Caddy log, mean the domain does not point at the server yet, or ports 80 and 443 are blocked. Check, in this order: DNS (`nslookup career.qualcsi.com` shows the Elastic IP), the AWS security group, UFW (`sudo ufw status`), then restart Caddy. Let's Encrypt retries on its own; wait a few minutes after fixing the cause.

N1. Start over completely (deletes all data on the server)

```
cd /opt/career-platform/deploy
docker compose -f docker-compose.prod.yml --env-file .env down
docker volume rm career-platform_pgdata
set -a; . ./env; set +a
docker compose -f docker-compose.prod.yml --env-file .env run --rm backend alembic upgrade head
docker compose -f docker-compose.prod.yml --env-file .env run --rm -e
ADMIN_EMAIL="$ADMIN_EMAIL" -e ADMIN_PASSWORD="$ADMIN_PASSWORD" -e ADMIN_NAME="$ADMIN_NAME"
backend python seed.py
```

N2. Restore a backup

```
cd /opt/career-platform/deploy
ls -lt backups/ | head
gunzip -c backups/leadgendb-.sql.gz | docker compose -f docker-compose.prod.yml --env-file .env
exec -T db psql -U postgres -d leadgendb
```

Replace `<date>` with the file name from the list. Restoring over a live database can cause errors if tables already exist. Ask for help before restoring on a server that has data you care about.

Part O. Security and demo rules

Before you share the address with anyone:

- **Use sample data only.** The enquiry form (for counselling requests) stores what visitors type. Ask testers to use fake names and numbers, or switch the form off.
- **Keep the admin password private.** Anyone with it can change the site's content. Change it after the demo: log in, then Admin → Users.
- **Keep the API documentation hidden if you do not want it public.** The page `/api/docs` lists every API operation. Ask for help to block it.
- **Never commit secrets.** The files `deploy/.env` and `deploy/backend.env` are excluded by the repository's ignore rules. Check with `git status` before you commit anything.
- **Keep the server updated.** Ubuntu installs security updates on its own (step 10). Update the platform with section M3 after each change.
- **Delete the demo when finished.** Stop the containers, stop the instance (section M5), then delete the instance and the Elastic IP, so no charges continue.

Known limits in this demo

- One server. If it stops, the site stops until it is started again.
- Visitors do not have student accounts yet. Only the admin panel has a login.
- Five collections point at ranking lists that do not exist in the database. Their colleges are shown in editor order, not ranking order.
- Search uses the database, not Meilisearch.

Appendix: Words used in this guide

Word	Meaning
AWS / EC2	Amazon Web Services. EC2 is its service for renting servers.
Instance	One rented server.
Elastic IP	A fixed internet address for your server.
Security group	The server's firewall rules, set in the AWS console.
SSH	A secure way to type commands on the server from your computer.
Key pair / .pem / deploy key	Files that prove who you are. Keep them private.
Terminal	The window where you type commands.
Docker / container	A packaged, isolated copy of an app. Each part of the platform is a container.
docker compose	The tool that starts several containers together, using <code>docker-compose.prod.yml</code> .
Environment file (.env)	A text file of settings and secrets read by the containers.
Let's Encrypt	The free certificate authority that gives the site its HTTPS padlock.
HTTPS / padlock	Encrypted website traffic. Caddy obtains the certificate from Let's Encrypt.
Migration	A script that changes the database structure (step 24).
Seed	Loading sample data into the database (step 25).
Swap	Disk space used as extra memory.